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Malpractice Liability Exposure and the Sports Medicine Team Physician

Caring for Professional Athletes in the National Football League, Major League Baseball, and National Hockey League

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Background: Orthopaedic surgeons play a critical role in ensuring the health and safety of professional athletes. Despite the privilege of treating elite athletes, there exists great financial exposure to individual physicians in the event of a malpractice lawsuit.

Hypothesis/Purpose: The purpose of this study was to evaluate and model malpractice liability exposure of the sports medicine surgeon caring for athletes in the National Football League (NFL), Major League Baseball (MLB), and National Hockey League (NHL) with respect to player position and additional supplemental malpractice insurance needs. It was hypothesized that routine liability coverage cannot adequately address the demands of caring for elite athletes in professional sports leagues.

Study Design: Economic and Decision Analysis; Level of evidence, 3.

Methods: In total, 2447 NFL, 992 MLB, and 980 NHL player contracts from the 2022-2023 season were aggregated from a publicly available online database. Position, team, total contract value, and mean yearly salary were noted. Risk ratios were calculated with respect to 1 million US dollars (USD) and 3 million USD of annual occurrence-based malpractice liability awards and used to generate a ''covered-to-treat'' analysis. Supplemental malpractice liability insurance was quantified.

Results: Assuming 1 million and 3 million USD occurrence-based awards covered by malpractice liability insurance, team physicians can fully treat 17.3% and 50.0% of NFL players, 43.2% and 59.7% of MLB players, and 13.6% and 41.0% of NHL players, without incurring additional personal financial risk, from a risk-based medicolegal model. Liability policies of 52.6 million, 108.1 million, and 64.1 million USD are required to treat 95% of NFL, MLB, and NHL players, respectively. Positions carrying the greatest risk ratios are quarterback (QB) (9.9) in the NFL, right field (15.1) in the MLB, and center (5.7) in the NHL.

Conclusion: Sports medicine specialists caring for elite athletes face potential personal financial risk due to insufficient medicolegal coverage. While coverage may vary among different practice settings including private, academic, or public state institutions, medical malpractice risk is crucial in partnerships between sports franchises, hospitals, players, agents, and physicians to protect sports medicine physicians and offer the highest-quality care.

Keywords: malpractice; liability; medicolegal; NFL

Recent public medical malpractice litigation has illuminated the liability that orthopaedic surgeons assume when caring for elite/professional athletes. Primary and consultant sports medicine surgeons willing to lend their time and expertise clearly absorb more risk when caring for individuals whose multimillion-dollar livelihood depends on their ability to return to peak athletic performance.[17,18] While malpractice litigation may be an area of prominent concern for sports medicine physicians, many physicians lack the legal expertise to comprehend their risk of exposure, or alternatively, consider that the responsibility of ensuring the health and safety of professional athletes supersedes the risk. Better collective understanding and bargaining is required to avoid devastating personal penalties, with some cases amounting to 30 million US dollars (USD) for the individual surgeon and 14 million USD for the involved groups.[9]

The recent understanding of chronic traumatic encephalopathy resulting from concussions and subclinical head trauma has fundamentally transformed major sports leagues, namely, the National Football League (NFL).[16] This transformation is most evidently seen by changes in medical management and gameplay rules aimed at

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safeguarding player health.[13] With increased recognition of chronic traumatic encephalopathy, the medical and social climate surrounding the NFL and other sports leagues has favorably shifted to prioritize player longevity over immediate on-field return to play, despite potential conflicts of interest regarding injury self-disclosure among players.[11] This change is exemplified by a recent notable malpractice case in which the prosecuting attorney stated, "This case and this jury may have changed the course of history by now forcing these team doctors and trainers to stop worrying about when a player might return to play and start thinking about the next 50 years of a player's life."[9,18] Despite lacking any vested clinical, financial, or moral interest in prematurely releasing a patient and elite player to rehabilitation and sport, sports medicine team physicians occasionally find themselves to be a focal point of the medicolegal judicial system when outcomes are unfavorable, even in circumstances beyond their control. One study found that orthopaedic surgeons across all subspecialties were nearly twice as likely to face medical malpractice claims compared with other specialties per year.[12] Furthermore, recent studies have found that malpractice liability has continued to rise in the past 3 decades, with some estimates of a 280% increase.[6]

Given the limited understanding surrounding the medical malpractice milieu in sports medicine literature, there is a critical gap in knowledge regarding the implications of malpractice liability for physicians in professional sports settings, particularly surrounding the care of teams whose players have premium contracts that likely surpass the team physicians' malpractice coverage limits. Varying contract structures across major leagues—such as guaranteed contracts in the Major League Baseball (MLB) and National Hockey League (NHL) versus the more conditional contracts in the NFL—further complicate the landscape.[7] Guaranteed contracts, which require teams to pay players irrespective of injury status or return to play, may mitigate the likelihood of legal disputes with players but could simultaneously increase pressure from key organizational stakeholders. In contrast, conditional contracts inherently heighten litigation stakes, as any perceived delay or mismanagement in return to play could result in significant financial repercussions for both the player and team. Given that the NFL has more players than other major sports leagues, as well as the highest revenue, the potential for such risk is further magnified.

While physician medical malpractice liability may pale in comparison to other team-based considerations, the escalating financial exposure to individual physicians in the event of a malpractice lawsuit can no longer be ignored. Protecting player interests and ensuring their smooth return to play must be balanced with the need to address physician security within this complex landscape. Therefore, the purpose of our study was to evaluate and model the malpractice liability exposure of the orthopaedic surgeon caring for athletes in the NFL, MLB, and NHL with respect to player position and additional supplemental malpractice insurance needs. We hypothesize that, even with conservative modeling, standard physician malpractice liability coverage is insufficient to adequately address the demands for caring for elite athletes in major sports leagues without support from professional franchises, affiliate hospitals, and external malpractice guardrails.

METHODS

Study Cohort

Contract data from NFL, MLB, and NHL players from the 2022-2023 season were identified from a publicly available source (https://www.spotrac.com/) on March 6, 2023.[20]

This data set included player position, team, total contract value, mean yearly salary, and total guaranteed money. NFL positions were grouped into 9 categories: quarterback (QB), wide receiver, tight end, linebacker, offensive lineman, defensive lineman, defensive back, running back, and kicker. NFL players of similar positions were grouped together to increase modeling potential as follows: center, left guard, right guard, left tackle, and right tackle were combined into offensive linemen; defensive edge and interior defensive linemen were combined into defensive linemen; safety and cornerback were combined into defensive backs; fullbacks and running backs were grouped into running backs; and punter, kicker, and long snapper were combined into kickers. MLB positions were grouped into the following 11 categories: first base, second base, third base, catcher, center field, designated hitter, left field, outfielder, pitcher, right field, and shortstop. Pitcher, relief pitcher, and starting pitcher were combined into the single category "pitcher." NHL positions were grouped into 5 categories and did not require combining: left wing, center, right wing, defenseman, goaltender. All active contracts were evaluated regardless of whether players were on active rosters, including injured players.

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Risk Ratio Modeling

Risk incurred by an orthopaedic surgeon for treating a professional athlete in the NFL, MLB, and NHL was modeled as a risk ratio. This was calculated by dividing the total contract value of a player by typical surgeon malpractice insurance. Policies included a liability limit of 1 million USD per claim up to a limit of 3 million USD per policy period. This policy was chosen as it is the most common liability insurance policy in the United States.[5] Occurrence-based awards provide coverage for incidents that occur during the policy period, regardless of when the claim is filed, offering long-term protection even after the policy has expired. In contrast, claims-based awards only cover incidents if both the occurrence and the filing of the claim happen within the policy period, making continuous coverage essential to avoid potential gaps in liability protection.[14]

The risk ratio model operates under 2 critical assumptions: (1) the orthopaedic surgeon treated a player immediately after signing their contract and (2) in the event of a malpractice lawsuit, the athlete sought damages equal to or exceeding the total value of their contract. Risk ratios are presented as absolute values. Importantly, the model does not factor lost wages from future contracts, endorsements, other ancillary revenue streams, or relevant factors beyond the monetary value of the players' contracts, including punitive damages and emotional costs.

Covered-to-Treat Player Analysis by Position

Professional athletes were stratified by their positions. The percentage of players for whom the team physician has sufficient malpractice coverage to treat ("covered-to-treat") was calculated as the number of players with risk ratios ≤ 1 divided by the total number of players for that position.

Supplemental Liability Insurance Analysis

The percentage of athletes that are covered-to-treat and total liability insurance were modeled up to 250 million USD. This value represents the proportion of athletes that a physician has sufficient malpractice coverage to treat before incurring personal losses. Using increasing intervals of 100,000 USD, we calculated the proportion as the number of players covered-to-treat divided by the total number of players in the respective league.

Statistical Analysis

Continuous data are presented as mean and standard deviation, while categorical data are presented as count and frequency. All data management and statistical analysis, including risk ratios, were performed using R Version 4.2.1 (R Core Team).

RESULTS

Cohort Characteristics

A total of 4419 contracts from 2447 NFL, 992 MLB, and 980 NHL players, which is also equivalent to the number of contracts in each league, were included from the 2022-2023 season. Within the NFL, there were 99 contracts for QBs, 447 contracts for defensive linemen, 440 contracts for offensive linemen, 305 contracts for wide receiver, 477 contracts for defensive backs, 157 contracts for tight end, 223 contracts for linebacker, 183 contracts for running backs, and 116 contracts for kickers. In the MLB, there were 38 contracts for first base, 47 contracts for second base, 55 contracts for third base, 75 contracts for catcher, 43 contracts for center field, 22 contracts for designated hitter, 40 contracts for left field, 38 contracts for outfielders, 532 contracts for pitcher, 32 contracts for right field, and 70 contracts for shortstop. In the NHL, there were 96 contracts for goaltenders, 315 contracts for defensemen, 299 contracts for centers, 157 contracts for left wings, and 113 contracts for right wings.

Contracts from NFL players varied by position, with QBs earning more than double other positions at 7.6 million USD per year. Designated hitters in the MLB earned the most at 8.2 million USD per year. In the NHL, left wings earned the most at 3.3 million USD per year. Table 1 illustrates the total contract value, total guaranteed salary, and annual mean salary for each position.

The distributions of contracts by position, which highlights the significant variation among players, is illustrated in Figure 1.

Risk Ratio Analysis

Assuming 1.0 as achieving maximum liability coverage for the sports medicine surgeon with a malpractice occurrence, the highest mean risk ratios based on a 3 million USD award were for QBs (9.9) in the NFL, right field (15.1) in the MLB, and center (5.7) in the NHL. Full risk ratios are explored in Table 2.

Covered-to-Treat Analysis

From a population-based perspective, the following 3 NFL positions have the lowest proportions of covered-to-treat risk ratios: QB, kickers, and defensive linemen. The following 3 MLB positions have the lowest proportion of covered-to-treat risk ratios: first base, right field, and designated hitter. Defensemen had the lowest proportion of covered-to-treat players in the NHL. The complete distributions are illustrated in Figure 2.

Supplemental Liability Insurance

Assuming 1 million and 3 million USD occurrence-based awards covered by malpractice liability insurance, NFL

TABLE 1
Summary of Player Contracts Stratified by Position[a]

Position	n	Total Contract Value, Million USD, Mean (Range)	APY, Million USD	Total Guaranteed Money, Million USD
NFL				
QB	99	29.6 (0.75-450)	7.6	16.5
DLM	447	11.2 (0.71-141)	3.5	5.4
OLM	440	10.6 (0.71-138)	3.2	4.8
WR	305	10.4 (0.71-140)	3.4	5.0
DB	477	8.4 (0.71-101)	2.7	3.7
TE	157	7.4 (0.71-75)	2.5	3.1
LB	223	7.0 (0.71-100)	2.3	2.9
RB	183	6.3 (0.71-90)	2.1	2.4
Kickers	116	6.3 (0.71-24)	2.1	1.8
MLB				
Right field	32	45.2 (0.72-365)	7.4	—
Designated hitter	22	43.3 (0.72-325)	8.2	—
Shortstop	70	39.8 (0.72-341)	6.1	—
3rd base	55	39.6 (0.72-350)	6.9	—
Center field	43	28.8 (0.72-427)	5.4	—
1st base	38	26.1 (0.72-225)	7.7	—
Left field	40	21.1 (0.72-189)	7.5	—
2nd base	47	21.1 (0.72-175)	5.0	—
Outfielders	38	19.2 (0.72-209)	3.4	—
Pitcher	532	11.9 (0.72-324)	4.3	—
Catcher	75	10.0 (0.72-116)	3.4	—
NHL				
Center	299	17.2	3.2	—
Right wing	113	15.2	3.1	—
Left wing	157	14.8	3.3	—
Defenseman	315	14.6	2.9	—
Goaltender	96	10.9	2.7	—

[a]APY, average per year; DB, defensive backs; DLM, defensive linemen; LB, linebacker; MLB, Major League Baseball; NFL, National Football League; NHL, National Hockey League; OLM, offensive linemen; QB, quarterback; RB, running backs; TE, tight end; USD, US dollars; WR, wide receiver. Guaranteed salary is clarified for the NFL but not applicable (denoted by a "—") for the MLB and NHL where contracts are fully guaranteed.

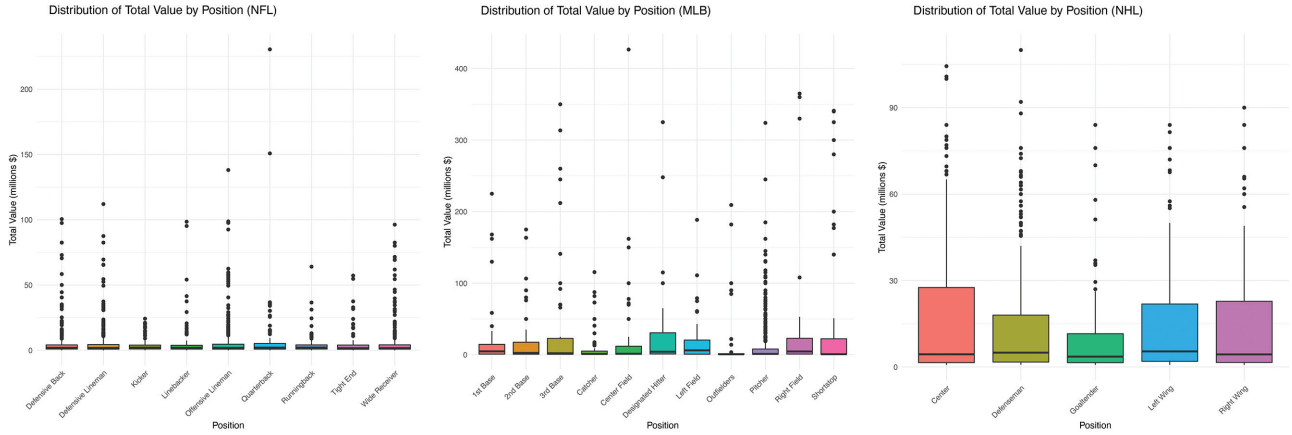

Figure 1. Distributions of contracts in terms of total value and average per year (APY) salary across the National Football League (NFL), Major League Baseball (MLB), and National Hockey League (NHL).

team physicians are only financially eligible to treat 17.3% and 50.0% of NFL players, respectively. Malpractice coverage totaling 52.6 million USD is required to treat 95% of NFL players. With those same thresholds of malpractice coverage, MLB team physicians are only eligible to treat 43.2% and 59.7% of MLB players, respectively. Malpractice

TABLE 2
Summary of Risk Ratios Stratified by Position, Including Mean (±SD) and Median (Minimum-Maximum) Ratios for a 1 and 3 Million USD Occurrence-Based Malpractice Policy[a]

		Mean RR ± SD		Median RR (range)	
Position	n	1 Million USD	3 Million USD	1 Million USD	3 Million USD
NFL					
QB	99	29.6 ± 70.7	9.9 ± 23.6	4.0 (0.75-450.0)	1.3 (0.25-150.0)
DLM	447	11.2 ± 21.4	3.7 ± 7.1	3.6 (0.71-141.0)	1.2 (0.24-47.0)
OLM	440	10.6 ± 19.3	3.5 ± 6.4	3.6 (0.71-138.1)	1.2 (0.24-46.0)
WR	305	10.4 ± 21.3	3.5 ± 7.1	2.4 (0.71-140.0)	0.8 (0.24-46.7)
DB	477	8.4 ± 16.3	2.8 ± 5.4	2.7 (0.71-100.5)	0.9 (0.24-33.5)
TE	157	7.4 ± 13.5	2.5 ± 4.5	2.6 (0.71-75.0)	0.9 (0.24-25.0)
LB	223	7.0 ± 15.2	2.3 ± 5.1	2.6 (0.75-100.0)	0.9 (0.25-33.3)
RB	183	6.3 ± 12.4	2.1 ± 4.1	3.2 (0.71-90.0)	1.1 (0.24-30.0)
Kickers	116	5.1 ± 5.8	1.7 ± 1.9	2.7 (0.71-24.3)	0.9 (0.24-8.1)
MLB					
Right field	32	45.2 ± 102.8	15.1 ± 34.3	4.6 (0.72-365.0)	1.5 (0.24-121.7)
Designated hitter	22	43.3 ± 85.9	14.4 ± 28.6	4.2 (0.72-325.0)	1.4 (0.24-108.3)
Shortstop	70	39.8 ± 89.1	13.3 ± 29.7	0.8 (0.72-341.0)	0.3 (0.24-113.7)
3rd base	55	39.6 ± 82.5	13.2 ± 27.5	2.1 (0.72-350.0)	0.7 (0.24-116.7)
Center field	43	28.8 ± 73.2	9.6 ± 24.4	1.4 (0.72-426.5)	0.5 (0.24-142.2)
1st base	38	26.1 ± 53.1	8.7 ± 17.7	4.8 (0.72-225.0)	1.6 (0.24-75.0)
Left field	40	21.1 ± 37.6	12.5 ± 2.1	6.2 (0.72-188.5)	2.1 (0.24-62.8)
2nd base	47	21.1 ± 53.1	7.02 ± 13.5	2.3 (0.72-225.0)	0.8 (0.24-53.3)
Outfielders	38	19.2 ± 49.0	6.4 ± 16.3	0.7 (0.70-209.3)	0.2 (0.24-58.3)
Pitcher	532	11.9 ± 29.6	4.0 ± 9.9	1.2 (0.70-324.0)	0.4 (0.23-108.0)
Catcher	75	10.0 ± 22.6	3.3 ± 7.5	1.2 (0.72-115.5)	0.4 (0.24-38.5)
NHL					
Center	299	17.2 ± 23.4	5.7 ± 7.8	4.4 (0.75-104.4)	1.5 (0.25-34.8)
Right wing	113	15.2 ± 20.8	5.1 ± 6.9	4.4 (0.75-90.0)	1.5 (0.25-30.0)
Left wing	157	14.8 ± 19.2	4.9 ± 6.4	5.4 (0.75-111.0)	1.8 (0.25-28.0)
Defenseman	315	14.6 ± 20.1	4.9 ± 6.7	5.0 (0.75-110.0)	1.7 (0.25-36.7)
Goaltender	96	10.9 ± 16.6	3.6 ± 5.5	3.6 (0.75-84.0)	1.2 (0.25-28.0)

[a]DB, defensive backs; DLM, defensive linemen; LB, linebacker; MLB, Major League Baseball; NFL, National Football League; NHL, National Hockey League; OLM, offensive linemen; QB, quarterback; RB, running backs; RR, risk ratio; TE, tight end; USD, US dollar; WR, wide receiver.

coverage totaling 108.1 million USD is required to treat 95% of MLB players. For the NHL, team physicians are only eligible to treat 13.6% and 41.0% of NHL players, respectively. Malpractice coverage totaling 64.1 million USD is required to treat 95% of NHL players. These percentages are illustrated in Figure 3.

DISCUSSION

Orthopaedic surgeons who care for elite athletes as primary team physicians or consultant experts are uniquely vulnerable for medical malpractice liability. Based on the most common occurrence-based coverage of 1 million and 3 million USD, and a conservative assessment of potential claims, our study finds that physicians are only financially prepared to care for 17.3% and 50.0% of NFL, 43.2% and 59.7% of MLB, and 13.6% and 41.0% of NHL players, leaving the great majority of team physicians at high risk in legal disputes. To attain 95% coverage, we found physicians that require malpractice coverages of 52.6 million, 108.1 million, and 64.1 million USD for the NFL, MLB, and NHL, respectively.

The tension between physicians and their commitment to prioritizing patient care and athletes' drive to return to peak performance—often intrinsically tied to their contracts—heightens the risk of physicians being held liable for medical malpractice. Recently, public litigation has called the interests and intentions of the team physician into question, which is further complicated by organizational interests and contractual commitments.[9] As a result, it is more imperative than ever for physicians to understand both the medicolegal terminology and climate of the respective communities in which they practice, including the limits of their respective insurance policies, the total value of their player-patient contracts, and any possible effects before providing care that has the potential to result in large financial consequences.

In the context of medical jurisprudence, it is essential to distinguish between maloccurrence and malpractice. Maloccurrence refers to an adverse outcome or complication that arises during medical care, which may not necessarily

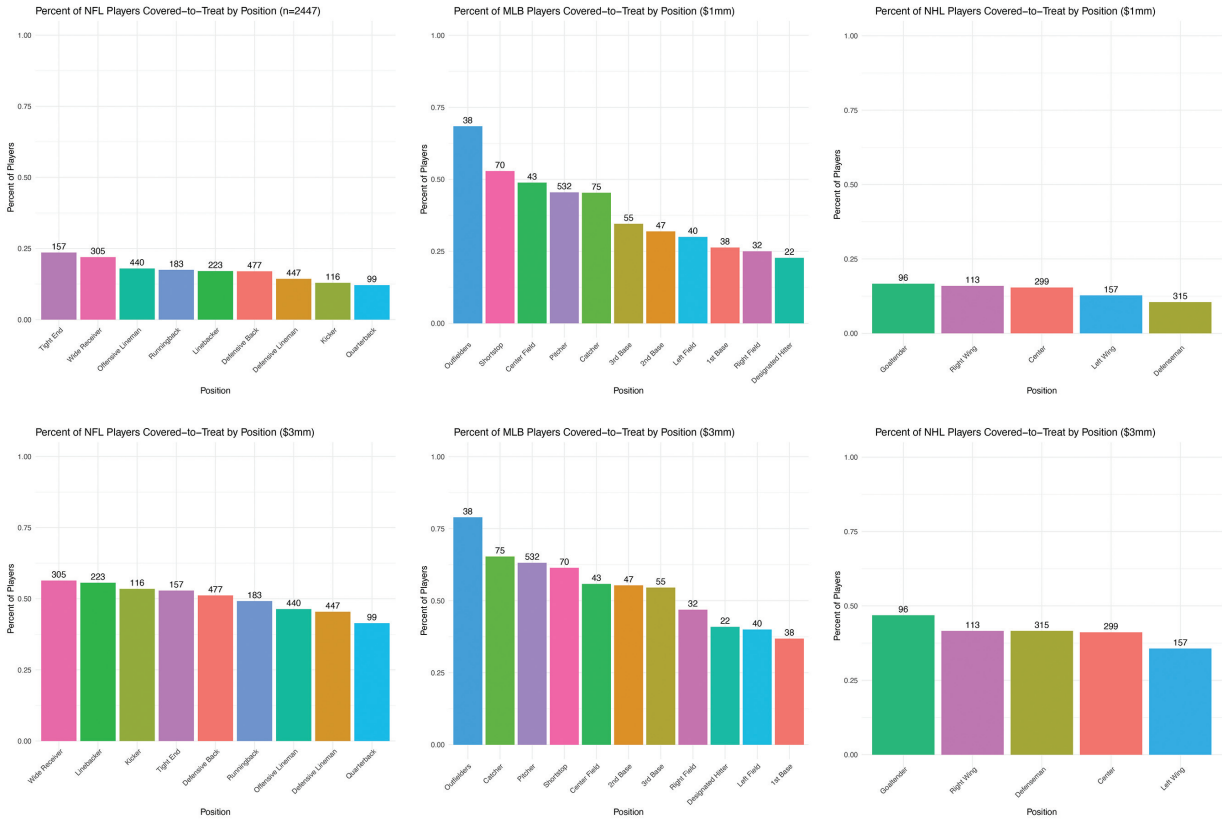

Figure 2. Distribution of covered-to-treat players by position across the National Football League (NFL), Major League Baseball (MLB), and National Hockey League (NHL) based on 1 and 3 million (mm) US dollar annual occurrence-based liability policy. The numeric values for each bar represents the total available players for that position; as an example, nearly 25% of the available 157 NFL TEs are covered-to-treat for a $1mm policy but just over 50% of the available 157 NFL TEs are covered-to-treat for a $3mm policy.

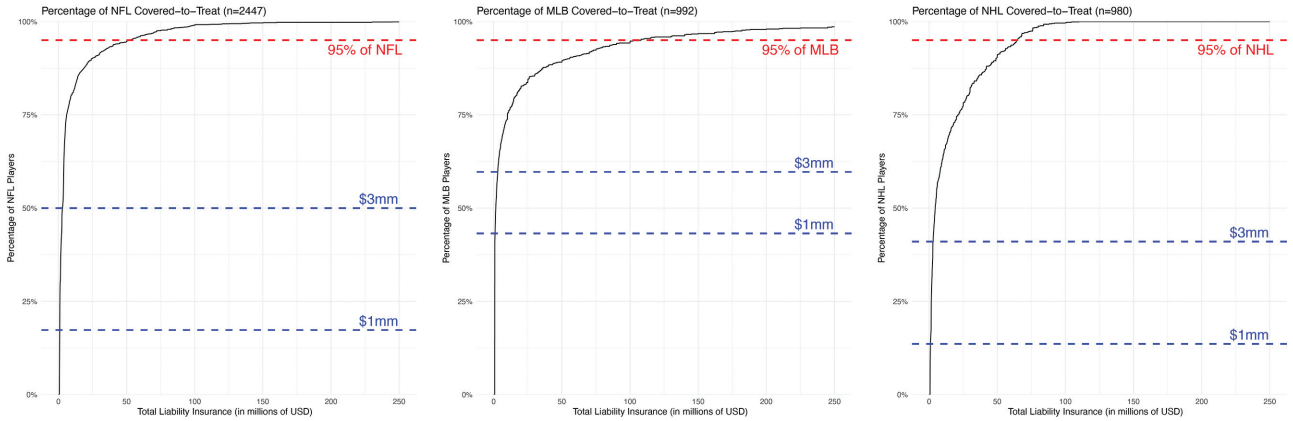

Figure 3. Distribution of requisite liability coverage to treat all players in the National Football League (NFL), Major League Baseball (MLB), and National Hockey League (NHL) from the 2022-2023 season. mm, million.

involve negligence or a breach of professional standards.[10] Conversely, malpractice is a specific legal term that encompasses professional negligence, wherein a health care provider fails to adhere to the established standard of care, resulting in harm to the patient.[19] The legal framework for establishing malpractice typically involves 4 critical components: (1) the existence of a professional duty owed to the patient, such as the duty to provide competent medical care; (2) a breach of that duty, in which the health care provider deviates from accepted medical practice; (3)

a causal link between the breach and the injury sustained by the patient; and (4) resulting damages, which may include physical, emotional, or financial harm.

Few studies have examined litigation exposure in sports medicine surgeons. Gatto et al[8] evaluated 328 orthopaedic sports medicine malpractice cases and found that 70.4% favored the physician, 22.9% favored the plaintiff, and 6.7% reached settlement. The most common reason for litigation was an intraoperative error (55.8%), with neurovascular injury resulting in the highest awards. Within spine literature, several studies have comprehensively evaluated malpractice litigation. In a query of 310 cases from the Westlaw legal database, Zhang et al[21] demonstrated that two-thirds of cases (67%) favored the physician, 24% favored the plaintiff, and 9% reached settlement, following similar trends to sports medicine. Once again, intraoperative error (28%) continued to remain the most common claim, followed by failure to obtain informed consent (24%). Although most legal proceedings favor physicians, they demonstrate the large burden of legal exposure that orthopaedic surgeons endure, including high legal costs and indemnity payments, potential reputational damage that may result from malpractice lawsuits from high-profile athletes, and notable emotional tolls that may lead to burnout and medical errors.[1] Furthermore, more than a quarter of all cases resulted in significant payments to the plaintiff. As litigation costs continue to increase, lawsuits have the potential to lead to significant financial damage to both surgeon and organizations.

These stakes are further amplified given the importance of functionality in the elite athlete treated by professional team physicians, and the profound effect of sports injuries on their quality of life. Medical malpractice is a specific tort in relation to professional negligence. Organized professional sports often require a physician to be present during contests.[15] The responsibilities for the team physician include assessing the athlete's medical eligibility to participate, which involves conducting physical examinations for new or backup players, as well as evaluating the eligibility of athletes recovering from injuries to resume athletic activities. They also must supervise any personnel providing health care services for athletes.[4] Ultimately, the team physician bears the responsibility for the entire sports medicine continuum, from determining clearance to participation to return-to-play decision, with each phase rigorously scrutinized.[2]

Despite the heavy legal risk, professional team physicians provide care for athletes often at the cost of great personal and financial consequences that are unlikely to be recouped by marketing benefits and remuneration. Therefore, to maintain the highest level of medical expertise, it is crucial for sports organizations to consider providing, at minimum, malpractice liability coverage that matches player contract valuations. However, this remains a contentious issue, as teams often argue against such coverage, citing concerns that it may encourage higher awards in lawsuits and questioning their obligation to provide coverage for team physicians, who are typically not direct employees. As of the 2022-2023 season, 1 million USD of annual occurrence-based awards—not claims-based awards—leaves team physicians at risk for awards that surpass their malpractice coverage relative to player contracts, effectively limiting their financial coverage to treat less than half of all players in the NFL, MLB, and NHL. This discrepancy is particularly pronounced for high-earning positions, with QBs and defensive linemen in the NFL, right field and designated hitters in the MLB, and centers in the NHL requiring supplemental coverage (Figure 1). Moreover, this issue extends beyond professional athletes, with significant implications for physicians treating nonathletes with high net worths, in which the potential liability may similarly exceed standard malpractice coverage, necessitating additional safeguards to mitigate financial risk.

Protection against malpractice lawsuits requires risk-sharing infrastructure to navigate medicolegal systems. Solutions to mitigating litigation risk include higher liability coverage, shared decision-making, and establishing a chain of care that works cohesively with the physician. While addressing organizational liability concerns, a sophisticated strategy that involves the input of consultants or other knowledgeable physicians in decision-making or that precludes any liability or additional coverage through a sharing agreement between the hospital and the franchise may reduce the risk of litigation for any one person. In addition, a chain of care must be formally reviewed by the physician in the initial contractual relationship with the professional sports team. In many cases, the chain of care, primarily in terms of rehabilitation, is not under physician control.[3] While physicians may argue that the details of rehabilitative aspects of care occur outside the direct involvement of the surgeon, the ultimate responsibility of activity clearance will rest on the team physician. Therefore, it is important to determine rehabilitation plans and athletic trainers and clarify roles and responsibilities before joining as a team physician. Any formal contract should stipulate that the team medical director has the sole authority to make all return-to-play decisions.

Still, no matter the risk sharing and review of contractual agreements, there continue to be outliers that result in an unfavorable equation for the physicians. This is best exemplified by the 15-fold risk ratio for treating right fielders in the MLB, assuming a generous 3 million USD. Thus, a comprehensive prerequisite malpractice liability plan is paramount. However, our findings are based on a conservative model that does not account for ancillary revenue streams or future contracts that contribute to net worth related to sport, nor does it account for punitive damages or pain and suffering that may affect the total monetary award. Professional sports organizations, the players' association for each respective league, physicians, and academic societies should work in tandem to develop comprehensive guidelines to address liability coverage and minimize litigation risk.

Limitations

As previously stated, the model is predicated on player contracts, which is an understated surrogate for a player's net worth. Moreover, no effort was made to project future

contracts, endorsements, and other revenue streams tied to player performance, which may minimize the risk team physicians experience. However, this risk is typically relegated to top players in the league who are outliers compared with our cohort of 2447 contracts analyzed. Player age was not considered, nor was the type of injury. Importantly, this model assumes every player will sue the treating physician. While malpractice lawsuits are rising, not all players are litigious. Moreover, individual and organizational liability policies pertaining to the team physician are unknown, accounting for our standard annual coverage plans. While we chose 1 million and 3 million USD as benchmarks of team physician liability coverage, many physicians and health systems opt for additional coverage through private entities. Given the private nature of these data, the exact proportion that our monetary benchmarks is applicable to physicians is unknown. Our findings were also limited to 3 major professional sports leagues and the context of malpractice lawsuits. Future research should include other professional sports organizations and international events, such as the National Basketball Association, Olympic events, and professional soccer leagues. Additionally, further investigation is warranted into collegiate athletics and medical liability under the National Collegiate Athletic Association, especially considering new policies like name, image, and likeness and their effect on care. The distinction between consulting and operating physicians should also be explored, as their roles carry different levels of legal risk. Lastly, mitigating factors such as thorough documentation and clear communication should be emphasized as strategies to prevent litigation.

CONCLUSION

Orthopaedic surgeons caring for elite athletes remain at great personal financial risk in the setting of insufficient medicolegal coverage. Our results suggest an urgent need for increased attention to malpractice risk in sports medicine. Even when excluding earnings from future contracts and ancillary sports-related endorsements, minimum malpractice coverages of 52.6 million, 108.1 million, and 64.1 million USD are prerequisite to accommodate 95% of NFL, MLB, and NHL contracts, respectively. Medical malpractice risk should be an increased source of focused attention when discussing partnerships between professional sports franchises, hospital organizations, players, their agents, and the treating sports medicine physician. Ensuring that physicians are adequately protected not only safeguards their well-being but also supports the health and safety of the athletes they serve.

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